

## MAHAJANGA, Madagascar

WMO: 670270

Lat: 15.667S

Lon: 46.352E

Elev: 87

StdP: 14.65

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 64.4       | 66.0 | 48.4                          | 50.5 | 84.9 | 51.3 | 56.2 | 82.7 | 21.2                  | 82.0 | 19.2 | 82.4 | 3.5                   | 160  | 0.375 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 11            | 15.1                   | 93.9            | 74.7 | 93.1 | 74.8 | 91.7 | 75.1 | 82.2                | 88.2 | 81.5 | 87.6 | 80.8 | 87.0 | 9.6                  | 100  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 80.6                            | 159.5 | 86.7 | 79.4 | 153.2 | 85.6 | 79.0 | 151.1 | 85.2 | 46.2          | 88.1 | 45.3 | 88.4 | 44.6 | 87.8 |                   |
| (p)                             |       |      |      |       |      |      |       |      |               |      |      |      |      |      |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |  |
| 18.7              | 16.5 | 14.8 | DB                         | 60.8 | 97.4               | 1.3 | 1.4                                                | 59.8 | 98.4       | 59.1 | 99.1       | 58.3 | 99.9       | 57.3 | 100.9 |  |
|                   |      |      | WB                         | 56.1 | 84.4               | 1.3 | 1.2                                                | 55.2 | 85.3       | 54.4 | 85.9       | 53.7 | 86.6       | 52.7 | 87.4  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 81.0   | 81.9  | 82.2  | 83.1  | 83.1  | 80.8  | 78.2  | 77.0  | 78.0  | 79.7  | 82.0  | 83.3  | 82.8  |
|                                                                                  | DBStd       | 2.88   | 2.02  | 1.82  | 1.69  | 1.64  | 2.13  | 2.13  | 1.88  | 1.83  | 1.84  | 2.04  | 1.88  | 2.02  |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDD50       | 11319  | 990   | 901   | 1026  | 993   | 956   | 845   | 837   | 869   | 892   | 993   | 1000  | 1017  |
|                                                                                  | CDD65       | 5844   | 525   | 481   | 561   | 543   | 491   | 395   | 372   | 404   | 442   | 528   | 550   | 552   |
|                                                                                  | CDH74       | 62380  | 5718  | 5312  | 6459  | 6254  | 5151  | 3682  | 3269  | 3674  | 4199  | 5721  | 6466  | 6475  |
|                                                                                  | CDH80       | 24968  | 2041  | 1927  | 2639  | 2739  | 2178  | 1448  | 1212  | 1448  | 1696  | 2379  | 2707  | 2554  |
| Wind                                                                             | WSAvg       | 7.1    | 7.2   | 6.6   | 6.4   | 5.6   | 5.9   | 6.5   | 7.3   | 8.1   | 8.3   | 8.1   | 8.2   | 6.8   |
| Precipitation                                                                    | PrecAvg     | 52.2   | 17.8  | 12.3  | 6.9   | 2.5   | 0.2   | 0.0   | 0.1   | 0.0   | 0.0   | 0.6   | 3.9   | 9.8   |
|                                                                                  | PrecMax     | 69.3   | 34.0  | 40.2  | 21.0  | 9.8   | 3.1   | 0.3   | 0.4   | 0.3   | 0.8   | 2.6   | 16.4  | 28.8  |
|                                                                                  | PrecMin     | 25.2   | 3.1   | 3.3   | 1.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
|                                                                                  | PrecStd     | 13.4   | 8.0   | 8.3   | 4.4   | 2.6   | 0.6   | 0.1   | 0.1   | 0.1   | 0.2   | 0.7   | 4.0   | 6.1   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 92.9  | 92.8  | 93.6  | 95.1  | 93.5  | 91.6  | 90.0  | 91.8  | 93.5  | 95.3  | 95.4  | 93.6  |
|                                                                                  |             | MCWB   | 77.5  | 78.7  | 77.9  | 77.2  | 74.1  | 71.7  | 69.3  | 68.9  | 69.8  | 71.8  | 73.6  | 77.0  |
|                                                                                  | 2%          | DB     | 91.0  | 90.6  | 92.3  | 93.3  | 91.8  | 89.9  | 89.2  | 90.0  | 91.6  | 93.3  | 93.2  | 91.7  |
|                                                                                  |             | MCWB   | 78.7  | 79.2  | 78.1  | 77.0  | 73.8  | 70.6  | 68.9  | 68.4  | 69.9  | 71.6  | 73.7  | 77.6  |
|                                                                                  | 5%          | DB     | 89.5  | 89.5  | 91.2  | 91.7  | 91.1  | 89.0  | 87.8  | 89.2  | 89.9  | 91.4  | 91.4  | 90.0  |
|                                                                                  |             | MCWB   | 79.4  | 79.5  | 78.4  | 77.0  | 73.7  | 70.2  | 68.4  | 68.2  | 69.7  | 72.3  | 75.3  | 78.4  |
|                                                                                  | 10%         | DB     | 87.9  | 88.0  | 89.7  | 90.9  | 89.5  | 87.6  | 86.3  | 87.6  | 88.1  | 89.6  | 89.8  | 89.2  |
|                                                                                  |             | MCWB   | 79.7  | 79.6  | 78.9  | 77.0  | 73.8  | 69.9  | 68.0  | 68.1  | 70.2  | 73.2  | 76.3  | 78.6  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 82.7  | 82.7  | 82.8  | 82.5  | 80.1  | 77.1  | 75.5  | 76.1  | 77.7  | 80.1  | 81.9  | 82.6  |
|                                                                                  |             | MCDB   | 88.2  | 87.8  | 88.8  | 89.4  | 87.4  | 83.7  | 81.8  | 82.6  | 84.7  | 87.5  | 88.1  | 88.9  |
|                                                                                  | 2%          | WB     | 81.6  | 81.7  | 81.9  | 81.0  | 78.9  | 75.6  | 73.8  | 74.5  | 76.1  | 78.9  | 80.3  | 81.3  |
|                                                                                  |             | MCDB   | 87.3  | 87.2  | 88.0  | 87.7  | 85.7  | 82.1  | 80.5  | 81.1  | 83.4  | 85.9  | 87.0  | 87.8  |
|                                                                                  | 5%          | WB     | 80.8  | 80.9  | 81.1  | 80.0  | 77.8  | 74.5  | 72.6  | 73.4  | 75.1  | 77.9  | 79.5  | 80.4  |
|                                                                                  |             | MCDB   | 86.6  | 86.6  | 87.2  | 86.6  | 84.7  | 81.3  | 80.0  | 80.3  | 82.2  | 84.9  | 86.2  | 86.8  |
|                                                                                  | 10%         | WB     | 80.0  | 80.1  | 80.2  | 79.2  | 76.7  | 73.3  | 71.5  | 72.3  | 74.1  | 77.0  | 78.9  | 79.8  |
|                                                                                  |             | MCDB   | 85.6  | 85.7  | 86.1  | 85.7  | 83.7  | 80.8  | 79.8  | 80.0  | 81.5  | 84.0  | 85.4  | 85.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.9  | 12.1  | 14.0  | 16.7  | 18.9  | 20.5  | 21.1  | 21.5  | 20.5  | 18.2  | 15.1  | 12.8  |
|                                                                                  |             | MCDBR  | 14.0  | 14.1  | 16.1  | 17.9  | 19.9  | 21.0  | 21.7  | 22.6  | 22.0  | 20.3  | 17.3  | 14.7  |
|                                                                                  |             | MCWBR  | 7.0   | 6.8   | 7.3   | 7.8   | 8.5   | 9.0   | 9.4   | 10.3  | 10.5  | 9.9   | 8.4   | 7.0   |
|                                                                                  | 5% WB       | MCDBR  | 12.4  | 12.5  | 13.7  | 15.9  | 18.0  | 19.7  | 20.6  | 20.8  | 19.8  | 16.7  | 13.9  | 12.8  |
|                                                                                  |             | MCWBR  | 7.1   | 7.0   | 7.3   | 7.8   | 8.8   | 9.5   | 10.1  | 10.7  | 10.3  | 8.7   | 7.3   | 6.9   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.424 | 0.414 | 0.387 | 0.358 | 0.343 | 0.326 | 0.327 | 0.340 | 0.376 | 0.417 | 0.452 | 0.427 |
|                                                                                  | taud        |        | 2.476 | 2.503 | 2.585 | 2.662 | 2.665 | 2.691 | 2.667 | 2.593 | 2.462 | 2.379 | 2.322 | 2.444 |
|                                                                                  | Ebn at Noon |        | 292   | 293   | 294   | 292   | 286   | 287   | 289   | 294   | 293   | 288   | 282   | 292   |
|                                                                                  | Edh at Noon |        | 37    | 36    | 32    | 28    | 26    | 25    | 26    | 30    | 36    | 41    | 43    | 39    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 1761  | 1829  | 1915  | 1866  | 1672  | 1563  | 1617  | 1850  | 2115  | 2256  | 2168  | 1968  |
|                                                                                  | RadStd      |        | 189   | 179   | 94    | 64    | 52    | 23    | 22    | 22    | 31    | 52    | 89    | 135   |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only           | +0.43 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page